Self-Sovereign Digital Certificate Verification Engine

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Traditional digital certificate verification often depends on centralized institutions or manual verification processes, which can make authentication time-consuming and difficult to manage at scale. This project proposes a Self-Sovereign Digital Certificate Verification Engine that provides a secure and decentralized approach for issuing, storing, and verifying digital certificates. The system allows an authorized institution to generate a digitally verifiable certificate containing relevant information about the certificate holder and issuing organization. Each certificate is associated with a unique digital identifier and a verification mechanism that allows its authenticity and status to be checked without relying solely on manual inspection. Certificate holders can maintain control over their digital credentials and share them with organizations when verification is required. The verification system checks the submitted certificate against the registered credential information and determines whether it is valid, altered, revoked, or unavailable in the verification record. The proposed system also provides an issuer interface for creating and managing certificates and a verification interface for organizations or individuals to validate submitted credentials. By reducing dependence on physical documents and manual verification, the system aims to improve the speed, reliability, and transparency of certificate verification. The project demonstrates how digital identity principles, secure credential management, and decentralized verification can be combined to provide a practical solution for trusted digital certification while giving certificate holders greater control over their credentials.

Index Terms: Certificate authentication, certificate verification, decentralized identity, digital credentials, digital certificates, digital identity, credential management, secure verification, self-sovereign identity, verifiable credentials